



SAFETY QUIZ: CHE381- Chemical Engineering Laboratory I

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

The most fundamental rule of chemical safety is awareness.

2. What two pieces of personal protective equipment must be worn at all times in the lab?

safety glasses and safety shoes

3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

lab coats, gloves, splash aprons, face shields, ear protection

4. True or False: Fume hoods are generally used as storage areas for chemicals.

False

5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

1000 ppm

6. What are the four routes of chemical entry into the body?

inhalation, injection, eye contact, skin contact

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

False

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8. What is the role of the Chemical Hygiene Plan?

to protect employees from health hazards associated with hazardous chemicals

9. What is the guideline for laboratory room airflow and fume hood airflow?

laboratory room airflow - 20 linear ft/min

fume hood - 80-120 linear ft/min

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

1.) keep aisles, hallways, and stairs clear

2.) keep work benches and work surfaces clear and organized

3.) surfaces and floors should be cleaned regularly

11. What is the proper method of transporting gas cylinders in the laboratory?

use 1 cylinder, 4 wheel carts

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

false

13. What is an accepted method of vapor detection in the laboratory?

draeger tubes, toxic gas monitors with alarms

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14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

- 1.) wear gloves
- 2.) wash promptly if skin comes in contact with chemicals
- 3.) waft chemicals

15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

transport with bucket type carrier designed to be shock resistant

16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

- 1.) evacuate immediate area
- 2.) contact facility operator, advise on nature of emergency and location
- 3.) operator will notify all employees to evacuate and activate emergency response team
- 4.) exit the lab, deactivate power
- 5.) evacuate the building

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17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

Flammable - flash point below 100°F

combustible - flash point between 100°F - 200°F

18. List the Emergency Procedure for an individual chemical exposure in the laboratory.



1.) remove person from exposure

2.) evacuate immediate area

3.) call 911 if necessary

4.) use eyewash / safety shower, flush for 15 mins

5.) consult SDS

6.) obtain medical evaluation

19. What is the definition of a "hazardous chemical"?

has an acute or chronic health effect may occur in cases of exposure

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?



safety showers - 30 gal/min should be checked monthly

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